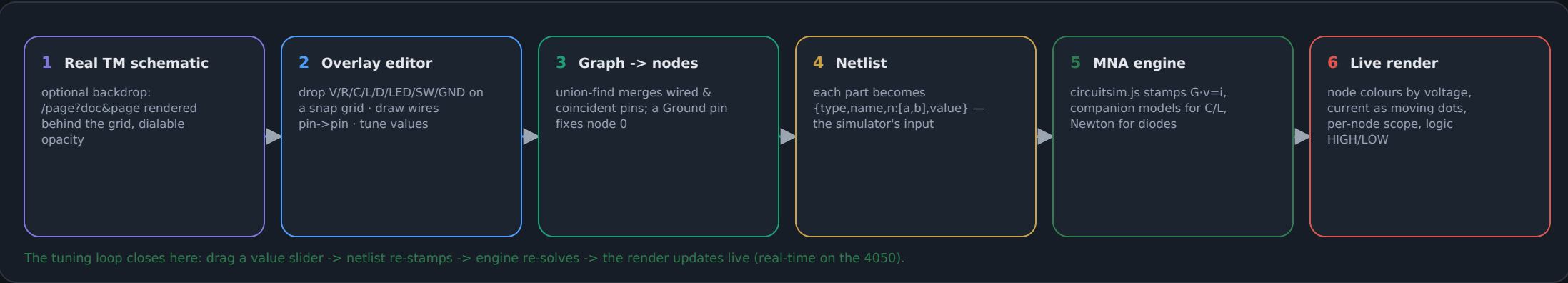


BUILT — Circuit Lab: overlay editor + real-time circuit simulator (v0.42.0)

Build or trace a circuit directly on top of a real TM schematic, then watch it run — a learning, advanced-viewing and display tool. 100% offline, custom MNA engine, no GPL.

1 · DATA FLOW — from a sheet to a live simulation



2 · THE EDITOR SURFACE

Tools

- Select — move parts, click to tune value (log sliders for R/C/L)
- Wire — click pin to pin, or to bare grid points
- Rotate (R) / Delete (Del) / Esc to cancel
- Place palette: Source, R, C, L, Diode, LED, Switch, Ground
- 6 demo circuits: divider, RC, RLC, rectifier, switch->lamp, logic

Run controls

- ▶ Run (animate transient) · DC (operating point) · Step
- Speed slider sets the timestep; Show = Analog or Logic view
- Scope: click any node to plot its voltage over time

3 · WHAT THE ENGINE SOLVES (validated)

Unit tests — all pass

● Voltage divider	2.500 V	exact
● Ohm's law	5.00 mA	exact
● RC transient @ tau	3.159 V	<1% vs 3.161
● Diode drop @ 4.4 mA	0.574 V	exact for $I_s=1e-12$
● Series D + LED	3.34 mA	converges (pnjlim)
● Underdamped RLC	overshoot->5 V	rings then settles

Backward-Euler companion models for C/L; SPICE-style voltage limiting lets series junctions and LEDs converge. Gaussian elimination w/ partial pivoting.

Dependency-free: runs as a browser global and as a Node module (how it's tested).

GROUNDING — a teaching overlay, not a claim about the manual (R1/R6)

The circuit is GROUNDING in what YOU build or trace: your placement is the source of truth — ideal for 'what happens if this relay coil opens, or this resistor doubles?'. The simulator never rewrites the TM, and the real cited sheet sits right behind the overlay. Auto-extracting a netlist straight from a raster TM (vision model) is deliberately NOT done here — that's the deferred power feature for the desktop/RTX 5070, and any such netlist would be labelled 'auto-extracted — verify against the TM' before use. Presentation-only and additive: the route is new, nothing existing changed.

RPS — Retroactive Post-Support degradation path

The live simulator is a modern-browser feature; on the 4050 it runs effortlessly. On the Win7/Vista LEGACY build it degrades gracefully: the editor still draws your circuit and the real schematic still opens in the static viewer — the animation simply turns off. So Circuit Lab is logged on the modern track from 0.42.0; the legacy track records it as 'static-overlay only (no live sim)'. Backwards-compatible and rollbackable (R1): delete the /circuitlab + /circuitsim.js routes and the two link buttons to fully revert.